

**TITLE OF THE PROJECT (FONT: TIMES NEW ROMAN,
FONT SIZE: 18)**

**A MAJOR PROJECT REPORT SUBMITTED (12)
IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE AWARD OF DEGREE OF**

MCA (14)

SUBMITTED BY (14)
Name of the Candidate(s) (14)
Roll Number(s): (14)



UNDER THE SUPERVISION OF(14)

Supervisor Name
Designation
Department of Computer Applications

SUBMITTED TO (font: 14)

Department of Computer Applications

Chandigarh Business School of Administration (Autonomous), Landran
(NAAC "A+" Grade Accredited & Affiliated with IKGPTU, Kapurthala, Punjab, India)

Mohali, Punjab, India

Year **(font: 16)**

CANDIDATE'S DECLARATION

I/We, **Name of the candidate(s), Roll Number(s)**, hereby declare that the work which is being presented in the major project entitled, “**Title of the project**” in partial fulfillment of requirement for the award of degree of **Master of Computer applications** and submitted in the Department of Computer Applications, Chandigarh Business School of Administration (Autonomous), Landran, Mohali, Punjab is an authentic record of my/our own work carried by me/us under the supervision of **Supervisor Name** (Designation, Institute) and **Co-Supervisor Name(if any)** (Designation, Department, Institute). The matter presented in this project report has not been submitted in this or any other University / Institute for the award of **Name of the program** Degree.

Signature of the Student

Dated:

(Name(s))

Roll Number(s)

Department of Computer Applications
Chandigarh Business School of Administration (Autonomous)
(NAAC “A+” Grade Accredited & Affiliated with IKGPTU, Kapurthala, Punjab, India)
Landran, Mohali, Punjab, India

Ref. No.:

Date:

CERTIFICATE

Certified that this major/minor project report entitled “.....**TITLE OF THE PROJECT.....**” is the Bonafide work of “.....**Name of the Candidate(S), Roll No.(s), of 4th Semester, MCA, Chandigarh Business School of Administration (Autonomous), Landran, Mohali, Punjab**”, who carried out the major project work under my/our supervision during January, 2026 - April, 2026 *(change accordingly)*.

(Name of Co-Supervisor)-if any

Co-Supervisor

Academic Designation

Department Name, CBSA

(Name of Supervisor)

Supervisor

Academic Designation

Department Name, CBSA

This is to certify that the above statement is correct to the best of our knowledge.

(Name)

Designation

HoD/Head PRC Committee

Department Name, CBSA

ACKNOWLEDGEMENTS

(It is just a format for hierarchy of the acknowledgement and can be extended/ reduced accordingly)

Minor/ Major Project work is an important aspect in the field of engineering, where contribution is made by many persons and organizations. The present shape of this work has come forth after contribution from different spheres.

Give acknowledgement to Co-Supervisor.

Give acknowledgement to Supervisor.

Give acknowledgement to H.O.D

Give acknowledgement to Director-Principal

I would also like to thank my parents, friends etc. who helped me in my Major Project.

I express my sincere gratitude to Chandigarh Business School of Administration (Autonomous), Landran, Mohali, Punjab for giving me/us the opportunity to work on the Major Project/ Minor Project during my/our final year of **MCA**.

At the end thanks to the Almighty for (entirely on personnel belief).....

(Name of the student)

Roll Number

ABSTRACT

The abstract is the precise of the complete work presented in the Major Project. It should be concise but at the same time should provide information about the work done and the methodologies used. The abstract may be written in the 1.5 line spacing paragraph format.

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Note 1: The number of Chapters is just indicative. There may be more number of Chapters in the Project Report but the format of the index has to be followed as given above.

Note 2: Page numbering should be as per the given format i.e. pages from candidates' declaration to list of abbreviations should be in Roman and page numbering of the Chapters should be in Arabic numerals. The title pages of the project report should NOT be numbered.

LIST OF TABLES(Font-14)

Table No.	Caption	Page No.
1.1	Caption of Table 1.1	3
1.2	Caption of Table 1.2	4
1.3	Caption of Table 1.3	5
2.1	Caption of Table 2.1	12
2.2	Caption of Table 2.2	13

All the table no. and captions should be in Time New Roman and of font size 12

LIST OF FIGURES

Figure No.	Caption	Page No.
1.1	Caption of Figure 1.1	3
1.2	Caption of Figure 1.2	4
1.3	Caption of Figure 1.3	5
2.1	Caption of Figure 2.1	12
2.2	Caption of Figure 2.2	13

All the figure no. and captions should be in Time New Roman, font size 12

ABBREVIATIONS USED

ANN	Artificial Neural Network
DRC	Departmental Research Committee
NFS	Neuro Fuzzy System
OC	Optical Communication

This list is optional. It should be included only if absolutely necessary, i.e. there being plenty of abbreviations that have been used in the Project. It should be contained in a single page if possible. The list should be in alphabetical order, Font size 12 in Times New Roman

- **Arrange in Alphabetical Order**

The Main Body of the Project

The main body of the project should comprise several chapters with the corresponding titles, and each page within these chapters must be numbered in numerals as page numbers. The usual way of presenting these chapters is given below.

Chapter 1: Introduction chapter. This chapter should contain brief background information about the project, the methodology implemented for problem solving and the outlines of the results and future scope of the project. It rarely contains drawings and graphical illustrations.

Chapter 2: Chapter of **Literature Review and Problem Outline**. It evaluates the current work with the previous one. It depicts the current implementations that overcome the previous problems and limitations of the project, and draws the attention and focus on the foreknowledge work that would be conducted based on the ongoing work at present, therefore, write **Problem Formulation** and **Objectives of the Project** followed by **Methodology**. At the end provide **Organization of the Report**. It must be clear and simple to understand.

Chapter 3-4 or 5: These chapters describe the overall in-depth information about the project. These chapters also involve the basic theoretical information about each and every component & aspect of the project, such as circuit design, simulation, implementation and modeling, software implementation, statistical analysis and calculations done, results gained, and so on.

The appropriate information should always be accompanied with pictorial representations, tabular demonstrations, diagrams, flow charts, visible graphs, Images, photos other representations and depictions of the project, along with simulation results with good resolution and clarity.

Results and Discussions chapter shall form the penultimate chapter of the project report and shall include a thorough evaluation of the investigation carried out and bring out the

contributions from the study. The discussion shall logically lead to inferences and conclusions as well as scope for possible further future work.

Conclusion and Future Scope

The conclusion and future scope part summarizes the whole report by highlighting all the chapters and their significance and the importance of the project and about the achievements.

This will be the final chapter of the project report. A brief report of the work carried out shall form the first part of the Chapter. Conclusions derived from the logical analysis presented in the Results and Discussions Chapter shall be presented and clearly enumerated, each point stated separately. Scope for future work should be stated lucidly in the last part of the chapter.

Referencing and Appendices

The project report must be considered as a very standard report, and therefore, it should follow all rules, guidelines and protocols of gathering and presenting information, and implementing that and drawing conclusions out of it.

All these activities require appropriate and authentic sources of information and that particular information must be referenced or cited according to the copyrights and other guidelines. Therefore, to make the report original, it should be free from plagiarism and must follow standard citations and guidelines of citations to represent the reference names.

The appendices of a project report should be written in Times New Roman format, and it should contain the information which is appropriate and added to the main text like Embedded C program code, raw data, and so on.

Organization of the Project Report

This project report shall be presented in a number of chapters, **starting with Introduction and ending with Summary and Conclusions**. Each of the other chapters will have a precise title reflecting the contents of the chapter. A chapter can be subdivided into sections, subsections and sub subsection so as to present the content discretely and with due emphasis. When the work comprises two or more mutually independent investigations, the project report may be divided into two or more parts, each with an appropriate title. However, the numbering of chapters will be continuous right through.

Report on the present investigation

The reporting on the investigation shall be presented in one or more chapters with appropriate chapter titles.

- Due importance shall be given to experimental setups, procedures adopted, Techniques developed, methodologies developed and adopted.
- While important derivations/formulae should normally be presented in the text of these chapters, extensive and long treatments, copious details and tedious information, detailed results in tabular and graphical forms may be presented in Appendices. Representative data in table and figures may, however, be included in appropriate chapters.
- Figures and tables should be presented immediately following their first mention in the text. Short tables and figures (say, less than half the writing area of the page) should be presented within the text, while large table and figures may be presented on separate pages.
- Equations should form separate lines with appropriate paragraph separation above and below the equation line, with equation numbers flushed to the right.

Chapter 1

TITLE OF CHAPTER 1

Please note that whatever be the content written in this template is just an example. In this chapter, scope of the work is first explained, and then the background and the proposed work are described. The problem formulation, objectives of Project, methodology, and the purpose of the work should also be discussed. Thereafter, a brief description of all the chapters is also given in organization of project report.

1.1 Title of Article 1.1

The “**Chapter 1**” should be written in font size 16 and the “**TITLE OF CHAPTER 1**” should be written in font size 16. Line spacing of the paragraphs should be 1.5 all through the text and do not miss the “one tab” space at the beginning of all the paragraphs. The headings of the articles should be in size 12 and the subheadings should be in size 12. All the headings should be in bold face.

1.2 Title of Article 1.2

The citations should appear as Ignizio [3, 4], Osyczka[6], Ignizio and Cavalier [5] and Badri et al. [1]. Note here that only the last names of the authors are being used. While citing a research article/book authored by more than 2 authors, “et al.” should be used with the last name of the first author so as to indicate the article/book is authored by more than 2 people but the names of all the authors should appear in the References section while describing the respective article/book. If more than one article by the same author(s) is being cited at a single place then the numbers of both the articles (separated by a comma) should appear in the same square bracket. Note Ignizio [3, 4] as cited above. If more than one citation by different authors is occurring together, then the citations should appear chronologically, i.e. the older article should be cited first.

1.3 Title of Article 1.3

The mathematical equations (if any) should be written using Equation Editor or Math Type only. All the equations should be centrally aligned and only those equations should be numbered which are being referred in the Project. Due care should also be given to the variables that appear in the running text. Such variables should also be written using the Equation Editor or Math Type using the inline equation option. It is also important to note that the bold face variables represent the vector quantities so the variables should not be written in bold face letters. Some examples are given below.

$$\sum_{i=1}^m x_{ij} \leq b_j \quad (j = 1, 2, \dots, n) \quad (1.1)$$

$$x_{ij} - y_j \leq 0 \quad (i = 1, 2, \dots, m; j = 1, 2, \dots, n) \quad (1.2)$$

Where x_{ij} represents the binary variable, y_j represents the variable assuming any real value. Also important to note the “small w” in “where” written after the equations to describe the variables present in the equations.

1.4 Title of Article 1.4

Sometimes a definition/comment is to be cited in the text. The same should be done as given below.

Osyczka [] has defined the multi-objective optimization as a problem of finding:

“A vector of decision variables which satisfies constraints and optimizes a vector function whose elements represent the objective functions. These functions form a mathematical description of performance criteria which are usually in conflict with each other. Hence, the term “optimizes” means finding such a solution which would give the values of all the objective functions acceptable to the designer.”

Note the single spacing in the above paragraph and the reduced margins in which the text appears.

1.5 Summary This will be last title of the chapter. Summarize the whole chapter here.

Chapter 2

TITLE OF CHAPTER 2 (LITERATURE SURVEY AND PROBLEM OUTLINE)

Note here that only the last names of the authors are being used. While citing a research article authored by more than 2 authors, “et al.” should be used with the last name of the first author so as to indicate the article is authored by more than 2 people but the names of all the authors should appear in the References section while describing the respective article.

Badri et al. [1] has discussed/proposed/presented/analyzed whatever main stating point of the paper, has to be mentioned. Then mention the research findings of the paper and new techniques used. Please go through the Abstract and introduction part thoroughly.

Note: Literature survey should be in past tense.

Nasimuddin et al. [17] **proposed a dual-band single-feed circularly polarized, L-shaped slotted patch antenna with a small frequency-ratio is for GPS applications. An L-shaped slot was cut at the centre of a square patch radiator for dual-band operation. A single microstrip feed-line was underneath the center of the coupling aperture ground-plane. The frequency-ratio of the antenna can be controlled by adjusting the -shaped slot arm lengths. The measured 10-dB return loss bandwidths for the lower and upper-bands were 16% (1.103–1.297 GHz) and 12.5% (1.444–1.636 GHz), respectively. The measured 3-dB axial-ratio (AR) bandwidth was 6.9% (1.195–1.128 GHz) for the lower-band and 0.6% (1.568–1.577 GHz) for the upper-band.**

Give brief description on gap analysis, it may be represented point wise.

Chapter 3

TITLE OF CHAPTER 3

3.1 Title of Article 3.1

3.2 Title of Article 3.2

3.2.1 Title of Article 3.2.1

The bullet marks may be used to list items. Try to use as simple bullet marks as possible as given below.

- List item 1
- List item 2
- List item 3
 - List item 3.1
 - List item 3.2
- List item 4

3.2.2 Title of Article 3.2.2

If some code/algorithm is to be given in the text, the same should be written in italics with “one tab” space and should be a single space paragraph. The most common example is written below

Code/Algorithm: Title Name

#include<iostream.h>

#include<conio.h>

```

void main()
{
    clrscr();

    cout<<"Hello World - This should bring some smiles";

    getch();
}

```

3.3 Title of Article 3.3

All the tables that are present in the Project report should be cited at the relevant places. The caption of the tables should be written above the table and should be written in font size 10 as given below in an example. While citing a table, complete “Table 3.1” should be written rather than “Tab. 3.1” or “table 3.1”.

Table 3.1: Example of Table

Col1	Col 2	Col3	Col4	Col5

The alignments of the columns may be adjusted as per the type of data being projected. More importantly, do not use the complete text width until unless it is required. The squeezed tables appear better than the other ones. Table 3.1 is an example of such a case. A table should preferably be completed in a single page, so the space should be adjusted accordingly. If however, a complete table cannot be put at the current position, then the running text should be continued there and the table should be provided at the top of the next page.

3.4 Title of Article 3.4

All the figures that are present in the Project report should be cited at the relevant places. The caption of the figures should be written below the figure and should be written

in font size 10 as given below in an example. While citing a figure, complete “Figure 3.1” should be written rather than “Fig. 3.1” or “figure 3.1”.

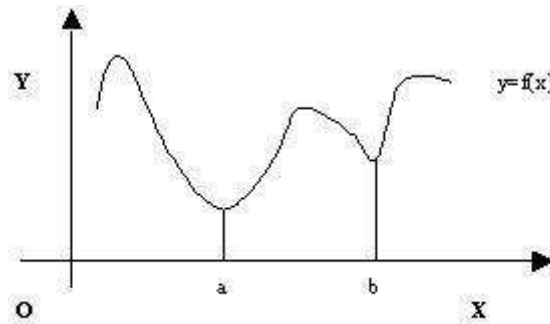


Figure 3.1: Example of Figure

3.5 Title of Article 3.5

In case of using an abbreviation and if the list of abbreviations has not been included in the Project report, the full form of the abbreviation should be mentioned at the first instance where it is introduced, thereafter the abbreviation can be used anywhere in the Project report. ANN (Artificial Neural Network) is a mathematical model that is inspired by the functional aspects of biological neural networks. In most cases, an ANN is an adaptive system that changes its structure based on external or internal information that flows through the network during the learning phase.

REFERENCES

The references should be sorted by year of publication in descending order. A few examples of the references that include research articles in magazines/journals or in conference proceedings, books, B.Tech/M.S. /Ph.D. Dissertation etc. are given below.

- Number all the references.
- Use a chronological bibliography.
- Each listed reference in the bibliography must be cited in the text of the report.
- For a book give the name(s) of author(s), title of book, edition, chapter number, and page numbers, publisher, location and year of publication.

Example:

[25] Jones, C.D., A.B. Smith, and E.F. Roberts, 1994, “Efficient Real-Time Fine Grained Concurrency”, 2nd Ed., Ch. 3, pp. 145-7, *Tata McGraw-Hill*, New Delhi.

For a journal/conference paper, give the name(s) of authors, title of paper, *name of journal/conference*, volume and issue number (for journal), page numbers, and month and year of publication.

Example:

[1] Badri, M.A., Mortagy, A.K. and Alsayed, A., 1998, “A multi-objective model for locating fire stations”, *European Journal of Operational Research*, Volume Number 110, Issue Number 2, pp. 243-260.

[2] Chatani, E., Hayashi, R., Lange, R. and Balny, C., 2002, “Thermal and pressure stability of Phe46 mutants of ribonuclease A”, *Proceedings of First International Conference on High Pressure Bioscience and Biotechnology*, Kyoto, Japan, pp. 27-32.

[3] Ignizio, J.P., 1968, “A Method to Achieve Optimum Air Defense Sensor Allocation”, *MS Dissertation*, University of Alabama, Alabama.

- [4] Ignizio, J.P., 1971, “A Heuristic Solution to Generalized Covering Problems”, *Ph.D. Dissertation*, Virginia Polytechnic Institute, Blacksburg.
- [5] Ignizio, J.P. and Cavalier, T.M., 1994, “Linear Programming”, *Prentice-Hall*, Englewood Cliffs, New Jersey, pp. 457-505.
- [6] Osyczka, A., 1985, “Multicriteria optimization for engineering design”, *Design Optimization*, Academic Press, Cambridge, pp. 193-227.

It is very important that all the references should be cited in the text and those cited in text should be mentioned in the references. Complete information of all the references should be provided, i.e. name of the journal, volume number, issue number, pp., year etc. The same goes for the books as well the details of the publisher should be provided here, however, the pp. in this case is optional. In case of a reference from a conference then the complete title of the conference should be provided together with the venue where it was held.

Appendix A (font size 16)

TITLE OF THE APPENDIX (font size 16)

Detailed information, lengthy derivations, raw experimental observations etc. are to be presented in the separate appendices, which shall be numbered in Roman Capitals.

Appendix B

TITLE OF THE APPENDIX

Appendix C

TITLE OF THE APPENDIX

Appendix D

TITLE OF THE APPENDIX (PUBLICATIONS)

Summary of the Report Format is Given as Below

- Report should be on A4 size paper.
- There should not be page number on Title page.
- Page numbering should be in roman (i, ii, iii....) for front pages and at left bottom on above mentioned and in numeric's (1, 2, 3....) and should be accordingly as used in this report i.e. in the footer at left bottom on chapter1 onwards.
- Write Project title in the footer.
- Margins: Top 1.0, Bottom 1.0, Left 1.5 and Right 1.0.
- Line spacing: 1.5 lines.
- Caption for table (size 10): - above the table, in bold only and rest simple and should be marked as Table 1.1, Table 1.2 (under chapter 1) and so on.

eg. Table 1.1: Title of the Table

- Caption for figures(size 10): - below the figure in bold only and rest simple and should be marked as Figure 2.1, Figure 2.2 (under chapter 2) and so on.

eg. Figure 1.1: Title of the Figure

- Font: - Times New Roman
- Size: - 12 for text, 12 for headings (sentence case, bold) 16 for Chapter no.(sentence case, bold), 16 for chapter title (Caps, Bold) and 12 for Sub headings (sentence case, bold)

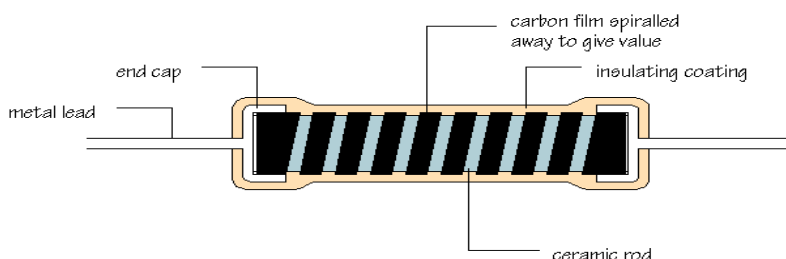


Figure 2.1: Title of the Figure

Table 1.1: Title of the Table

INPUT	OUTPUT
1	0

PROJECT FORMAT

REPORT

Paper Quality

The project report shall be printed on white bond paper, whiteness 95% or above, weight 70 gram or more per square meter.

Size

The size of the paper shall be standard A 4; height 297 mm, width 210 mm.

Chapter and Section Format

Chapter

Each chapter shall begin on a fresh page.

Sections and Subsections

A chapter can be divided into Sections, Subsections and Sub subsections so as to present different concepts separately. Sections and subsections can be numbered using decimal points, e.g. 2.2 for the second section in Chapter 2 and 2.3.4 for the fourth Subsection in third Section of Chapter 2. Chapters, Sections and Subsections shall be included in the contents with page numbers flushed to the right.

Table / Figure Format

Small size table and figures (less than half of writing area of a page) should be incorporated within the text, while larger ones may be presented on separate pages. Table and figures shall be numbered chapter wise.

For example, the fourth figure in chapter 5 will bear the number Figure 5.4.

Reference for Table and Figures reproduced from elsewhere shall be cited in the last and separate line in the table and figure caption, e.g. (Name of the table [12]).

Binding

This is the final and delicate stage of the art of report writing and preparation. As simple as it looks, it can spoil all the efforts that have been put forth on the project report, if handled carelessly. The final hardbound project report should look professional and neat, since this is the first point of contact with the reader. An aesthetically pleasing cover page of a report can surely persuade the reader to go deeper into the contents of the report. This could only be achieved if the student hands the job to a competent printer.

Lettering

All lettering on Title cover (Maroon) shall be embossed in gold.

Bound back

The degree, the name of the candidate and the year of submission shall also be embossed on the bound (side) in gold.

Blank Sheets

In addition to the white sheets (binding requirement) two white sheets shall be put at the beginning and the end of the project report.